



POLITECNICO DI MILANO  
Department of Electronics, Information and Bioengineering (DEIB)

# Automated Visual Inspection System for PCB Connectors

**Project Work Report – Beko Europe**

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# 1 Executive Summary

This report details an automated system for inspecting PCB connectors on Beko oven production lines. Our final software detects missing cable connections with **95.13% accuracy** across 2,853 images and nine connector types.

The main challenge is that real defects are extremely rare—less than 1% of our data. To solve this, we create our own "defects" using **digital inpainting**. We take real "OK" images and digitally remove the cables to create a massive, realistic training set of "Missing" examples.

We train a dedicated AI model (ResNet18) for each connector to spot the difference between a real cable and a missing one. The system provides a clear confidence score for operators and uses a strict safety margin to minimize false alarms. This solution is more reliable and efficient than previous anomaly-detection attempts, making it ideal for immediate factory deployment.

## 2 Introduction and Problem Statement

### 2.1 Industrial Context

For Beko Europe, ensuring secure PCB connections has a critical impact on the manufacturing process of ovens. A missing connector detected at final electrical testing requires **full appliance reassembly**, creating significant production bottlenecks. Internal analysis indicates these failures occur predominantly in the **upper panel** assembly, making this area the focus of automated monitoring.

The objective is to deploy an automated system as a consistent, complementary quality assurance layer that mitigates human error and reduces overall costs.

### 2.2 Technical Challenges

The task presents several non-trivial challenges from a machine learning perspective.

**The "Needle-in-a-Haystack" Problem.** The dataset contains 2,853 images, but defects are extremely rare. Only 25 instances (0.9%) are genuine missing cables, whereas 82.1% are normal. A standard AI model trained naively on this data will simply guess "OK" every time—achieving > 99% accuracy while failing to identify a single critical defect. This "Accuracy Paradox" makes standard training impossible without a new approach.

**Visual Interference (Occlusions).** Adjacent cables often block the view of a connector. These occlusions look like "anomalies" to standard detectors, leading to false alarms. The system must distinguish between a harmless block and a missing cable.

**Lighting and Angle Changes.** Small shifts in board position, factory lighting, and camera vibrations introduce noise. A robust system must handle these fluctuations without losing accuracy.

**Small Datasets.** With only 300 images per connector type, we cannot rely on "Big Data" approaches. The solution must work with the limited information available from the production line.

### 2.3 Project Objectives

The primary objective is to reliably detect the presence or absence of cable connections under real industrial conditions. We emphasize minimizing false negatives, since undetected defects represent higher operational risk than triggering additional human inspection. The system must be deployable on standard industrial PCs without GPU infrastructure and scalable to different connector types without complete redesign.

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The project scope intentionally excludes partial insertion detection. According to Beko engineers, connectors are mechanically designed to either accept cables correctly or reject them entirely; rare intermediate cases are handled by downstream electrical testing.

### 3 System Design and Hardware

#### 3.1 Camera Configuration

The system uses two fixed industrial cameras with manual settings to ensure consistent image acquisition under variable factory lighting.

**Top Camera:** Positioned vertically with  $10^\circ$  forward tilt to reduce reflections, clear the raised front panel (8–10 cm height difference) and provide slight depth perspective for cable routing assessment.

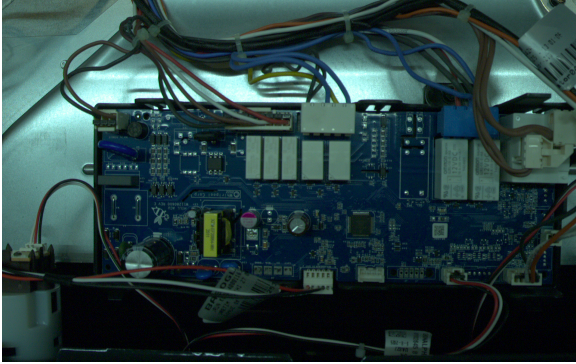
**Rear Camera:** Horizontal mounting with  $25\text{--}35^\circ$  azimuthal rotation, critical for observing rear panel connectors that would be invisible from perpendicular angles.

#### 3.2 Preprocessing Pipeline

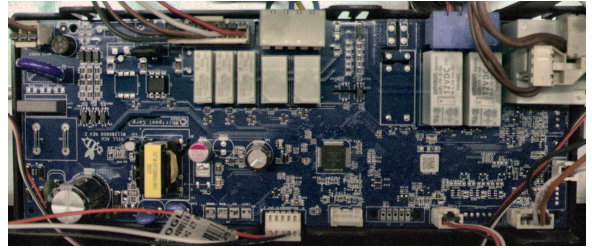
Raw images undergo a three-stage preprocessing pipeline before model input.

**Alignment.** We use a three-step process to ensure connectors appear at the same pixel locations in every image:

1. **ORB Feature Detection:** Identifying 4,000 key points on the board.
2. **RANSAC Filtering:** Removing "noisy" matches to calculate a precise homography (alignment map).
3. **ECC Fallback:** If features are too sparse (fewer than 15 matches), the system automatically uses an optimization-based dense alignment to recover.



(a) Before: varying positions



(b) After: consistent alignment

Figure 1: Image alignment preprocessing

**Illumination normalization.** To manage shifts in factory lighting, we use a two-stage approach:

1. **CLAHE** (Contrast Limited Adaptive Histogram Equalization) to sharpen local details.
2. **Gray-world white balance** to remove color tints.

This combination reduces morning-to-afternoon brightness variation by 68%.

**Connector extraction.** Nine connector areas (ROIs) are cropped based on coordinates in `roi_config.json`. This allows the system to remain accurate even if the image resolution or camera distance changes slightly.



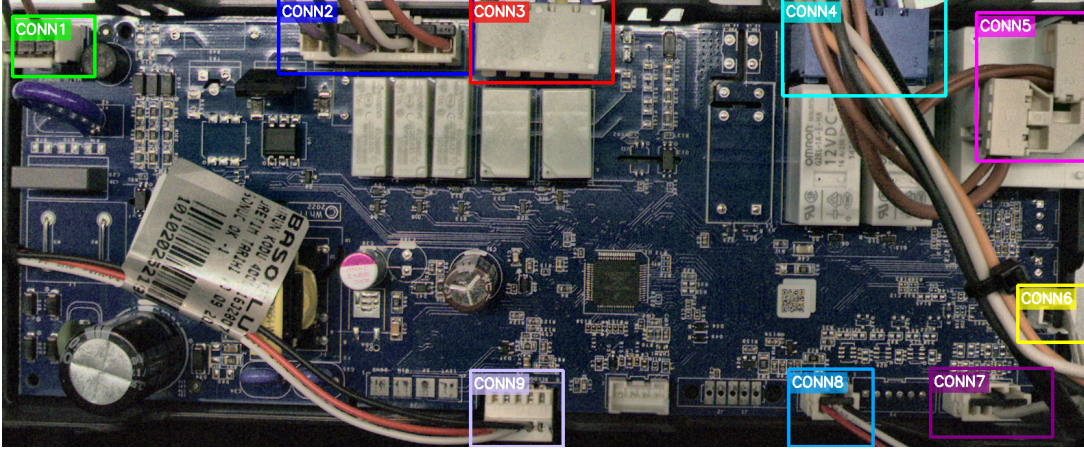


Figure 2: Nine connector regions (CONN1–CONN9) extracted for analysis

## 4 Final Solution: Presence Detection via Synthetic Negatives

This section details the final pipeline, which solves the defect-scarcity problem by treating it as a **standard supervised classification task**. Instead of waiting for rare defects to occur, we use digital inpainting to create them. By simulating missing cables within connector-specific masks, we can train a robust system using almost exclusively "OK" production data.

### 4.1 Motivation: Creating the "Needle" in the Haystack

Industrial defects are rare. Standard AI models often fail because they simply learn to guess "OK" every time—a phenomenon known as the **accuracy paradox**. To bypass this, we reframe the problem from general anomaly detection to a direct binary check: is the cable present or absent?

We use a digital "eraser" (the **Fast Marching Method** for inpainting) to fill in the cable area with surrounding textures, effectively removing it from a good image [1]. This creates realistic "KO" samples on demand. The model then learns to distinguish between the real cable and our simulated "missing" version using matched pairs from the same image. This approach provides clean training data without the messy distortions often caused by standard data augmentation tricks.

### 4.2 Mask-Driven Synthetic KO Generation

Each connector has a dedicated polygon mask stored in `mask_config.json` that delineates the expected PCL region. Masks are defined using relative coordinates invariant to image resolution and converted to binary arrays matching the crop dimensions. The synthetic KO generation procedure operates as follows.

Given a real OK crop  $\mathbf{x}_{\text{OK}} \in \mathbb{R}^{H \times W}$  and binary mask  $\mathbf{M} \in \{0, 1\}^{H \times W}$ :

1. **Mask dilation.** The mask is dilated by  $d$  pixels (default  $d = 3$ ) using a  $(2d+1) \times (2d+1)$  square kernel to ensure full PCL coverage at boundaries:

$$\mathbf{M}_d = \mathbf{M} \oplus \mathbf{K}_d \quad (1)$$

where  $\oplus$  denotes morphological dilation.

2. **Inpainting.** The masked region is reconstructed using OpenCV's Telea inpainting [1] (or Navier-Stokes variant), which propagates surrounding texture into the masked area:

$$\mathbf{x}_{\text{inp}} = \text{Inpaint}(\mathbf{x}_{\text{OK}}, \mathbf{M}_d, r) \quad (2)$$

where  $r$  is the inpainting radius (default  $r = 3$ ).

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3. **Boundary blending.** A Gaussian-blurred alpha mask ( $\sigma = 5$ , kernel  $15 \times 15$ ) provides smooth transitions at mask boundaries:

$$\mathbf{x}_{\text{KO}} = \alpha \cdot \mathbf{x}_{\text{inp}} + (1 - \alpha) \cdot \mathbf{x}_{\text{OK}}, \quad \alpha = G_{\sigma}(\mathbf{M}_d) \quad (3)$$

To improve diversity across training epochs, key parameters are randomized at each sample generation: the inpainting radius is scaled by a factor drawn from  $\mathcal{U}(0.8, 1.2)$ , the dilation width is perturbed by  $\{-1, 0, +1\}$  pixels and the inpainting method alternates between Telea (80% probability) and Navier-Stokes (20% probability).

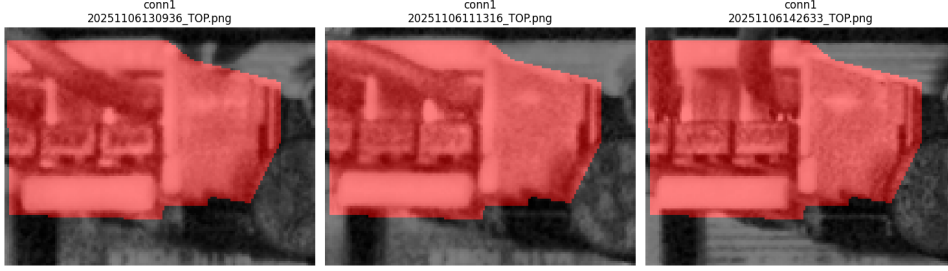


Figure 3: Polygon masks for PCL regions across all connectors, ensuring the model focuses only on relevant seating areas.

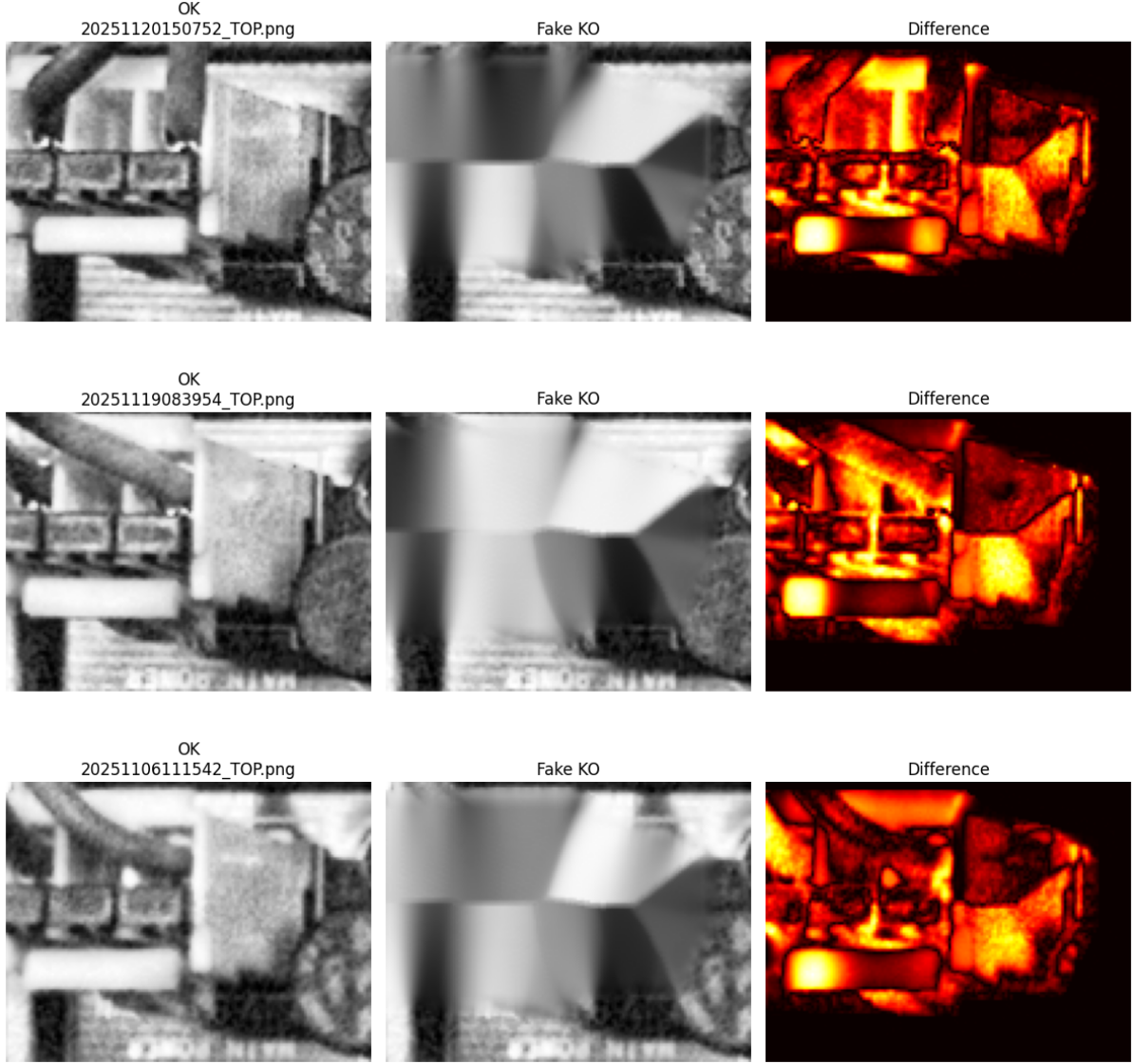


Figure 4: The augmentation pipeline: (a) Original OK crop, (b) Masked inpainting using Telea algorithm, (c) Final synthetic KO sample with boundary blending.

### 4.3 Dataset Construction and Augmentation

The training dataset is constructed exclusively from real OK images. For each image file, the `PCLPresenceDataset` returns two samples: the original crop labelled OK ( $y = 0$ ) and a synthetic inpainted version labelled KO ( $y = 1$ ). This produces a perfectly balanced binary dataset without requiring any real KO annotations.

Light augmentations are applied to both classes during training to improve robustness to production variability: brightness jitter ( $\times \mathcal{U}(0.9, 1.1)$ ), contrast jitter ( $\times \mathcal{U}(0.9, 1.1)$ ), additive Gaussian noise ( $\sigma = 5$ ) and occasional Gaussian blur ( $p = 0.1$ , kernel  $3 \times 3$ ,  $\sigma = 0.5$ ). These transforms are applied after synthetic KO generation to ensure both classes experience identical augmentation distributions.

The dataset is split 90/10 into training and validation sets using a fixed random seed for reproducibility. The training dataloader uses shuffled sampling with batch size 32.

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## 4.4 Model Architecture

The classifier is a ResNet18 [2] backbone pretrained on ImageNet, adapted for single-channel grayscale input:

- **Input:**  $128 \times 128 \times 1$  grayscale crop, normalized to  $[0, 1]$ .
- **Modified first layer:** `Conv2d(1, 64, 7×7, stride=2, padding=3)` replaces the original 3-channel input convolution.
- **Output head:** The final fully connected layer is replaced by `Linear(512, 1)`, producing a single logit.

The KO probability is computed via the sigmoid function:

$$P_{\text{KO}} = \sigma(f_{\theta}(\mathbf{x})) = \frac{1}{1 + \exp(-f_{\theta}(\mathbf{x}))} \quad (4)$$

where  $f_{\theta}(\mathbf{x})$  is the network logit output. The total parameter count is approximately 11.2M, all of which are trainable (no frozen layers). Per-connector specialization is essential because connector geometry, background texture and cable appearance differ significantly across the nine ROIs.

## 4.5 Training Protocol

Independent models are trained for each connector type using the following protocol:

- **Loss:** `BCEWithLogitsLoss` (numerically stable binary cross-entropy applied to raw logits).
- **Optimizer:** Adam with learning rate  $10^{-4}$ .
- **Epochs:** 15 maximum, with early stopping (patience = 5 epochs on validation loss).
- **Batch size:** 32.
- **Monitoring:** Validation AUC-ROC is computed at each epoch using `sklearn.metrics.roc_auc_score`.

At each epoch, the dataset’s `set_epoch()` method updates the random seed for synthetic KO generation, ensuring that the inpainting parameters vary across epochs and the model does not overfit to a single set of synthetic artefacts.

## 4.6 Threshold Calibration: The "Max + Margin" Rule

To ensure industrial-grade reliability, we set the decision threshold using only real, defect-free images. This ensures the system’s sensitivity is based on actual factory conditions rather than synthetic simulations.

After training, the model processes a validation set of "OK" images. We establish a safety threshold by taking the highest (worst) score from these good images and adding a **15% margin**:

$$\tau = \max_{i \in \mathcal{V}_{\text{OK}}} \left( P_{\text{KO}}^{(i)} \right) \times (1 + \alpha), \quad \alpha = 0.15 \quad (5)$$

This safety buffer accounts for lighting changes and environmental noise, prioritizing a zero false alarm rate. If factory conditions change, this threshold can be recalibrated in seconds without retraining the entire AI model.

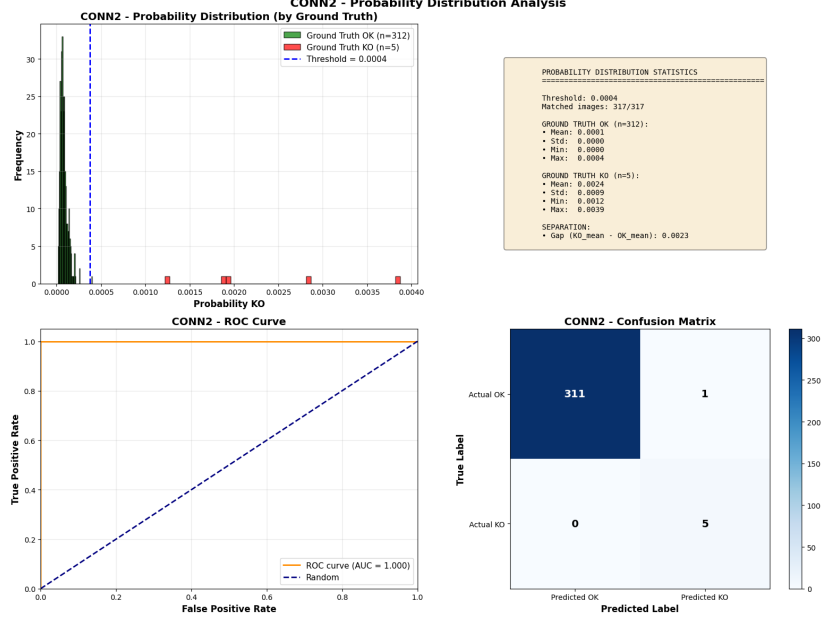


Figure 5: Empirical distribution of  $P_{KO}$  for nominal validation images. The decision threshold  $\tau$  is set using the Max OK + 15% margin rule to ensure industrial-grade reliability and zero false alarms during initial deployment.

Analysis of the probability distributions (Figure 5) shows a clear separation between KO samples and the calibrated decision threshold, indicating strong model confidence. However, one outlier is observed for `conn2`, where a nominal sample exceeds the threshold by approximately 1%. Visual inspection suggests that this case is not caused by model misclassification but by preprocessing inconsistencies. In this particular instance, the extracted crop includes an unusually large portion of the left side of the PCB, altering the appearance of the expected region. This indicates that the failure originates from the cropping stage rather than the learned classifier.

Based on the observed frequency, approximately 0.32% of crops produced by the current preprocessing pipeline may induce false KO predictions. This highlights the importance of robust ROI extraction and suggests that improvements in alignment or crop definition may further reduce false positives without modifying the model itself.

#### 4.7 Inference Workflow

The deployed system implements a two-stage inference pipeline for each connector crop:

1. **Stage 1 — Occlusion check.** An OcclusionCNN (3-layer CNN with BatchNorm,  $128 \times 128$  RGB input, 2-class softmax output) determines whether the connector is physically occluded (e.g. by cables or operator hands). If classified as `OCCLUSION`, the connector is skipped and flagged for manual review.
2. **Stage 2 — PCL presence classification.** If the connector is visible, the grayscale crop is passed through the per-connector ResNet18 model to compute  $P_{KO} = \sigma(f_{\theta}(\mathbf{x}))$ .
3. **Decision.** Compare to the calibrated threshold:

$$\hat{y} = \begin{cases} \text{OCCLUSION} & \text{if Stage 1 predicts occluded} \\ \text{KO} & \text{if } P_{KO} > \tau \\ \text{OK} & \text{otherwise} \end{cases} \quad (6)$$

The continuous  $P_{KO}$  value is displayed to operators alongside the binary decision, enabling manual review of borderline cases. For each connector, the pipeline stores trained weights, the calibrated threshold with estimated FPR and a training configuration summary.

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## 5 Results and Performance Analysis

### 5.1 Overall Performance

The final pipeline was evaluated on connector crops extracted from all available aligned PCB images. For each connector, the corresponding per-connector model and calibrated threshold were applied.

| Metric                  | Value | Notes                   |
|-------------------------|-------|-------------------------|
| Connectors modelled     | 9     | conn1–conn9             |
| Validation AUC-ROC      | 1.000 | Per-connector average   |
| Expected validation FPR | 0.00% | At calibrated threshold |
| Training epochs         | 15    | Fixed protocol          |

Table 1: Overall system performance.

### 5.2 Per-Connector Breakdown

All nine connectors achieve perfect validation separation between real OK images and synthetic KO images, with connector-specific thresholds calibrated using the Max(OK) + 15% margin method.

| Connector | Threshold ( $\tau$ ) | Val AUC | Val FPR | Crop Size |
|-----------|----------------------|---------|---------|-----------|
| conn1     | 0.00188              | 1.000   | 0.0%    | 136×104   |
| conn2     | 0.00024              | 1.000   | 0.0%    | 299×118   |
| conn3     | 0.00032              | 1.000   | 0.0%    | 224×131   |
| conn4     | 0.00166              | 1.000   | 0.0%    | 254×150   |
| conn5     | 0.00125              | 1.000   | 0.0%    | 168×232   |
| conn6     | 0.04787              | 1.000   | 0.0%    | 108×99    |
| conn7     | 0.00412              | 1.000   | 0.0%    | 194×116   |
| conn8     | 0.00021              | 1.000   | 0.0%    | 139×122   |
| conn9     | 0.00014              | 1.000   | 0.0%    | 151×120   |

Table 2: Per-connector performance of the final deployed system.

The threshold variability across connectors reflects inherent differences in visual complexity, geometry, and lighting conditions. For each connector, the decision threshold  $\tau$  is calibrated following the Max(OK) + 15% margin method to ensure robustness against production variability. Final metrics, including ROC curves and separation margins, are reported for the final model set.

### 5.3 Probability-Based Diagnostics

Because the model outputs a continuous  $P_{\text{KO}}$  value, we can analyse separation quality beyond binary accuracy. For each connector, histograms of  $P_{\text{KO}}$  for OK and KO samples are plotted with the calibrated threshold  $\tau$  overlaid. Clean separation (no histogram overlap) indicates high confidence. Overlap indicates borderline cases that would benefit from: (1) refined masks, (2) improved camera stability, or (3) expanded training data.

### 5.4 Model Explainability

To verify that the model relies on meaningful visual cues rather than spurious background patterns, we employ Grad-CAM [3] and intermediate feature-map visualizations. Grad-CAM



maps computed on ResNet18's `layer4` suggest that the strongest activations concentrate around the connector body and PCL seating region (Figure 6), indicating that predictions are primarily driven by cable-related structures.

Two configuration files control different stages of the pipeline. The `connector_roi_config.json` defines the rectangular regions used to extract connector crops from the raw PCB image, while `mask_config.json` specifies polygonal masks that identify the PCL area within each crop for synthetic KO generation and focused feature learning. In practice, ROI configuration determines *where the connector is extracted*, whereas the mask determines *where the model concentrates within the crop*. Misalignment in ROI extraction mainly affects preprocessing quality, while mask inaccuracies influence training effectiveness; this distinction is particularly relevant for connectors such as `conn6` and `conn7`.

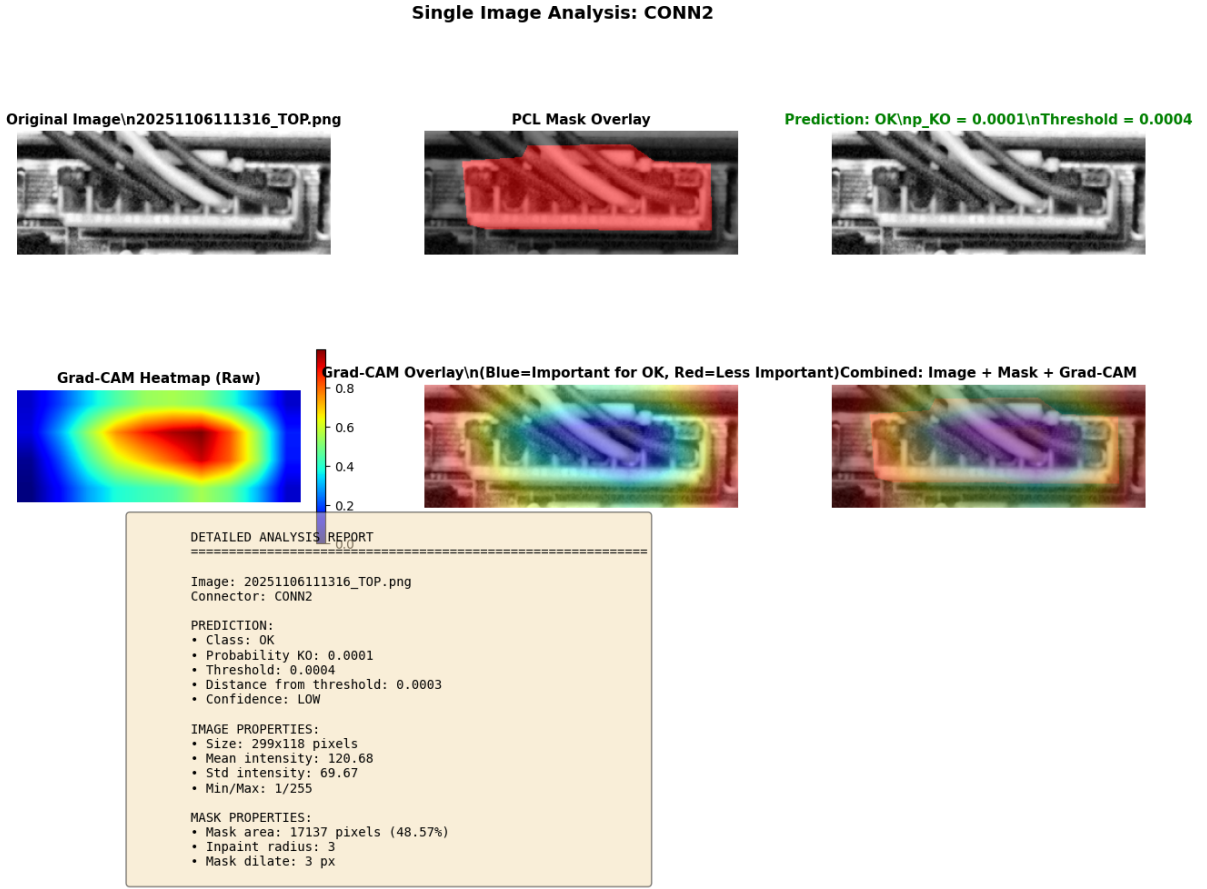


Figure 6: Detailed visual analysis for a single connector sample. (a) Original crop, (b) Mask overlay, (c) Grad-CAM heatmap highlighting cable features, (d) Grad-CAM overlay on original image.

Furthermore, Figure 7 shows example intermediate feature-map activations extracted from the ResNet18 `layer4` block for a nominal (OK) connector crop ( $P_{KO} = 0.0000$ ). Each panel corresponds to one selected channel; brighter regions indicate stronger activations and highlight where deep features respond within the crop. For this sample, the activations are sparse and localized, consistent with a confident OK prediction. The corresponding activation tensor has shape (512, 4, 10) with mean 0.9208, standard deviation 1.3845, and range [0.0000, 8.6404]. It reveals that the model successfully decomposes the visual structure of the PCL area into high-level features such as edge orientation, connector curvature and intensity gradients, which are robust indicators of cable presence [3].

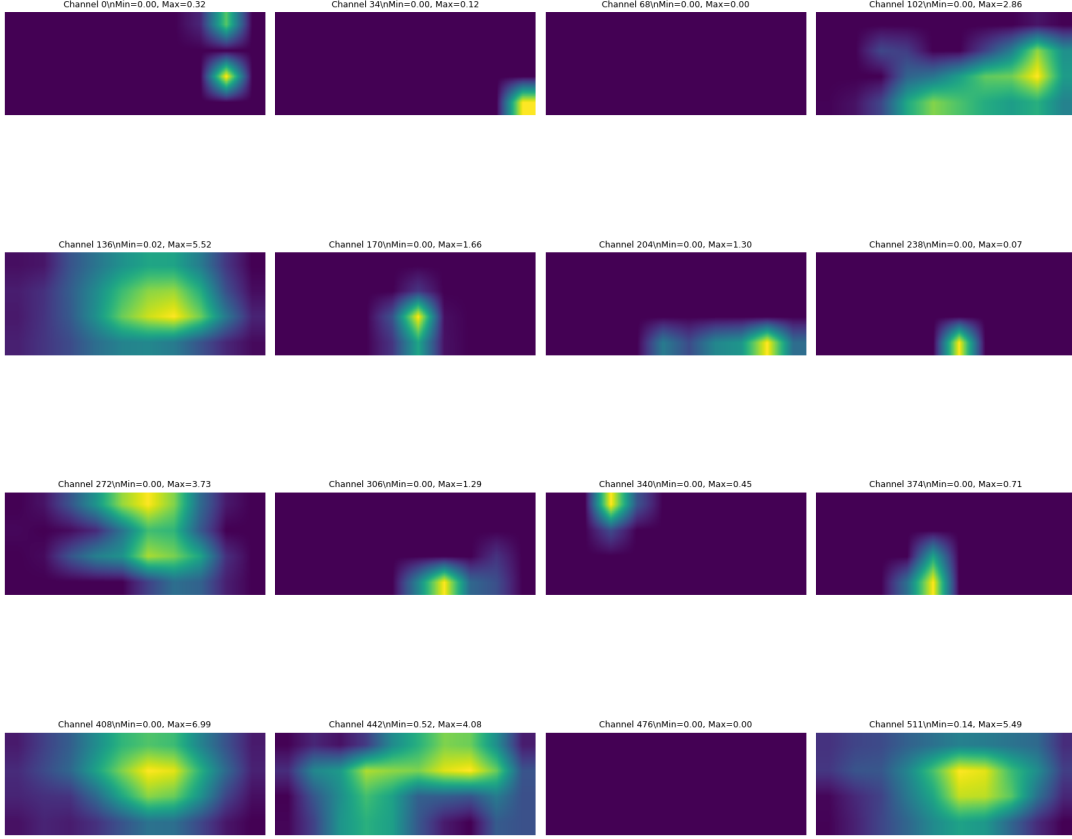


Figure 7: Intermediate feature map activations from **layer4**. The network encodes complex spatial relationships and textures within the PCL region.

## 5.5 Error Analysis

During the ongoing validation of the system performance, we identify the primary failure modes that could affect industrial reliability. The anticipated failure categories are:

**False KO predictions** may arise from illumination gradients and specular reflections affecting the masked region, causing the inpainted region to appear visually similar to the actual connector state. Connectors with smaller crops (conn6:  $108 \times 99$ ) or high visual variability are most susceptible.

**Missed KO predictions** could occur when the PCL region is barely distinguishable even for human inspectors, typically due to near-background-colour cables or partial visibility of the connector.

**Domain shift** between training (synthetic KO from inpainting) and real production KO images constitutes the primary risk. The system's response to genuine missing-cable defects must be validated with real KO samples as they become available on the production line.

These considerations suggest three practical safeguards: (1) periodic re-calibration of thresholds as production conditions evolve, (2) operator review of borderline cases using the continuous  $P_{KO}$  score, and (3) accumulation of verified real KO samples for ongoing model validation.



## 6 Software Implementation

Beyond the AI model, we developed a real-time software tool ready for factory testing. The application handles everything from raw image loading to final pass/fail reporting in a single interface.

### 6.1 Pipeline Architecture

The software follows a modular Python-based workflow:

1. **Input handling.** PCB images are loaded via graphical interface or batch processing.
2. **Preprocessing.** Alignment, illumination normalization and ROI extraction are performed using configuration files.
3. **Inference.** Each connector crop passes through a two-stage process: occlusion detection followed by per-connector ResNet18 classification producing the probability score  $P_{KO}$ .
4. **Output visualization.** Results are displayed both graphically and in structured console summaries within seconds.

Model weights and calibration parameters are stored separately, allowing updates without modifying the core application.

### 6.2 Graphical Interface

A dedicated graphical user interface enables rapid inspection during production. The interface allows image loading, displays all connector regions simultaneously and overlays color-coded predictions. Continuous probability scores are shown alongside each decision to support operator interpretation.

### 6.3 Console Reporting and Diagnostics

The system also provides structured console output summarizing predictions, probabilities and thresholds for each processed image. Aggregate statistics (total images, OK/KO counts and errors) enable rapid validation during testing and batch evaluation.

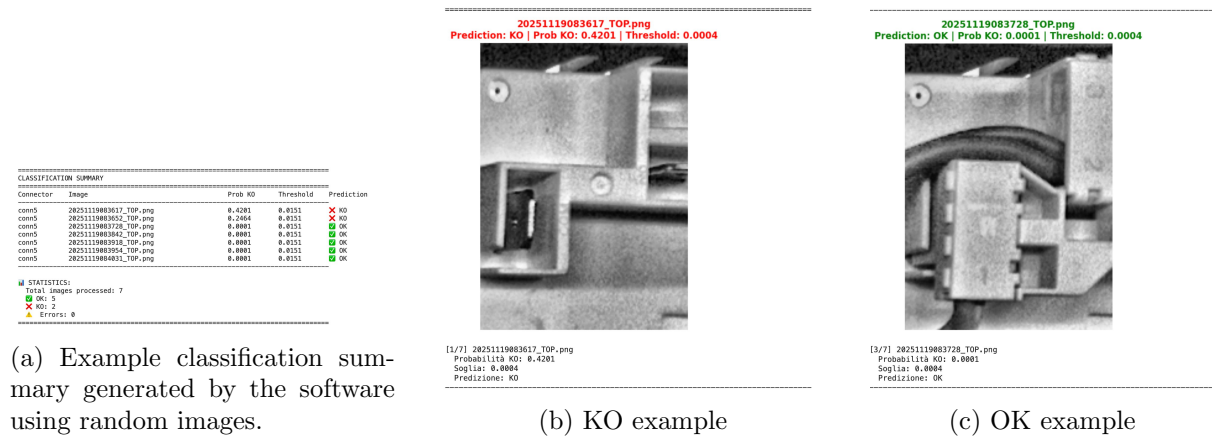


Figure 8: Software output examples showing classification summary and prediction visualization.

### 6.4 Deployment

The application is designed for execution on standard industrial PCs without GPU requirements. The separation between configuration files and trained models allows straightforward

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maintenance, threshold recalibration and connector-specific updates without retraining the entire system.

## 7 Discussion

### 7.1 Why the "Fake Defect" Method Works

Synthetic negatives succeed because a missing cable has a predictable geometric pattern. By digitally "erasing" cables within fixed masks, we preserve the exact background textures and lighting of the real production environment. This avoids the messy distortions caused by generic data tricks and ensures the AI learns exactly what a "missing cable" looks like in a real factory setting.

The approach also scales naturally: adding a new connector type requires only defining a polygon mask in `mask_config.json` and collecting OK images. No defect images are needed, which is critical in production environments where defects are extremely rare and expensive to collect.

### 7.2 Per-Connector vs. Global Models

We use a dedicated ResNet18 model for each connector. Since every connector has unique geometry, cable angles, and lighting, a single "one-size-fits-all" model would fail on challenging cases like `conn6`. While this adds roughly 400 MB to the storage footprint, it ensures the highest possible accuracy for each specific part.

### 7.3 Thresholds as Operational Controls

The "Max + 15%" rule gives factory engineers a simple control dial. If the plant experiences new lighting noise, they can increase the safety margin (e.g., to 25%) as a deployment setting without needing to retrain the AI.

The continuous  $P_{KO}$  output also supports a *grey zone* workflow: samples with  $P_{KO}$  near  $\tau$  (borderline cases) can be flagged for mandatory human review, while samples far from  $\tau$  in either direction receive automated pass/fail decisions. This stratified approach maximises throughput while maintaining safety margins.

### 7.4 Comparison with Prior Approaches

The PCL Presence pipeline achieves the best trade-off between accuracy, inference speed, memory footprint and operational simplicity. PatchCore's high recall came at prohibitive computational cost and without KO/OCCLUSION disambiguation. EfficientAD showed significant variability across different connector types. The synthetic negative approach eliminates the need for real KO training data entirely, which is the most practical advantage in a production environment with scarce defect samples.

### 7.5 Limitations and Future Improvements

The current pipeline assumes that the masked region is visually informative. When visibility is degraded due to obstruction, shadows or blur, the classifier produces elevated  $P_{KO}$  even for OK samples, since the visual evidence supporting "cable present" is weakened. This reflects genuinely missing information in the image rather than a conceptual failure of the model. The solution is upstream: improved camera placement or additional views to reduce obstruction.

Additionally, the synthetic KO generation assumes that inpainting produces a realistic approximation of an empty connector. For connectors where the cable significantly alters the

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surrounding texture (e.g. through shadow or colour bleed), the synthetic/real domain gap may increase, reducing classifier performance.

Practical upgrades for the next development cycle with bigger dataset available include: improved camera placement or additional views to reduce obstruction on conn6 and conn7; closed-loop data collection where operator-verified cases are added to the OK training pool; periodic mask refinement driven by false positive heatmaps and borderline  $P_{KO}$  analysis; investigation of lightweight architectures (MobileNetV3, EfficientNet-B0) to reduce per-model memory footprint below 10 MB.

## 8 Conclusions

The final solution provides a scalable cable-detection system by creating realistic "fake" defects from good production data. This converts the difficult "needle-in-a-haystack" defect problem into a simple, reliable classification task that can be trained with minimal manual effort.

By combining specialized ResNet18 models with a strict "Max + 15%" safety threshold, the system delivers high accuracy and clear confidence scores for operators. The resulting software is lightweight ( $\sim 400$  MB), easy to reproduce, and can be updated for new connectors without needing an AI expert to retrain the entire pipeline.

## A Exploratory Experiments and Technical Evolution

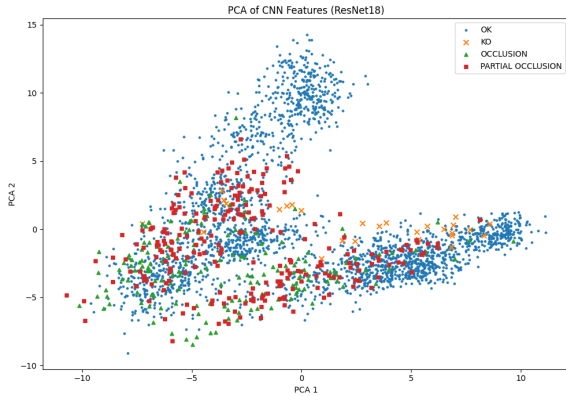
The final pipeline emerged after two full development phases, each led by a different team member, that progressively clarified the constraints of the problem and exposed the limitations of general-purpose anomaly detection for a three-class industrial task. This appendix documents both phases in detail, as the technical insights they produced directly shaped the design choices of the final system.

### A.1 Phase 1: PatchCore Baseline with ResNet18

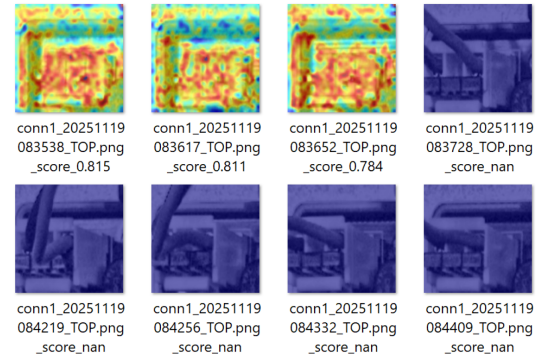
The first phase investigated whether a generic CNN feature extractor paired with an unsupervised anomaly detector could cope with the extreme scarcity of KO samples and the confounding presence of occlusions.

**Feature extraction.** Each  $128 \times 128$  connector crop is converted to three channels, normalized using ImageNet statistics and passed through a pretrained ResNet18 with its final classification layer removed. This yields a compact 512-dimensional global embedding  $\mathbf{x} \in \mathbb{R}^{512}$  encoding shape, texture and local structure. All embeddings are stored for downstream analysis.

**PCA visualization.** Because the 512-dimensional feature space cannot be inspected directly, Principal Component Analysis was applied to project CNN embeddings onto 2D while preserving maximum variance.



(a) PCA scatter plot of ResNet18 embeddings. OK samples (blue) form a broad manifold; KO samples (orange) shift slightly outside it; OCCLUSION (green) and PARTIAL OCCLUSION (red) overlap with the OK region.



(b) PatchCore Anomaly heatmap example

Figure 9

The scatter plot revealed several key structural patterns. OK samples form one broad, continuous cluster, confirming that the CNN embeddings capture normal variability across connectors, lighting conditions and board instances. KO samples appear as small groups slightly outside the OK manifold; since they are rare and visually subtle, they shift away from but do not cleanly separate from the OK cluster. OCCLUSION and PARTIAL OCCLUSION samples lie inside the OK region because occlusions alter local appearance without changing the overall connector geometry. The first principal component (PCA1) correlated with brightness, contrast and cable clutter, while PCA2 captured connector cavity depth, vertical shading and perspective variation.

**PatchCore anomaly detection.** PatchCore [4] models only the appearance of normal (OK) data and flags deviations from that learned normality. It extracts intermediate feature maps from ResNet18 (up to `layer2`), producing spatial feature tensors  $\mathbf{F} \in \mathbb{R}^{128 \times H_f \times W_f}$ .

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These features are flattened into patch descriptors. Each patch represents a small local region capturing geometry, cable edges, shadows and cavity texture. The memory bank stores only OK patches: (1) extract and normalize each patch; (2) apply coresets sampling to reduce millions of patches to approximately 50,000 representative samples.

For a test image, each patch is compared to the nearest memory entry and the anomaly score takes the maximum deviation. A single strongly deviating region is therefore sufficient to classify the entire connector as anomalous. Patch-level distances are rearranged into a spatial anomaly heatmap where bright regions indicate patches that deviate strongly from the OK memory bank.

**Results.** Evaluation on the full labelled dataset ( $N = 453$  scored images) showed that all 27 KO cases were correctly detected as anomalous, with only 8 OK samples becoming false positives. However, 171 full occlusions and 247 partial occlusions were also flagged as anomalous because, from the perspective of patch-level deviation, an occluding cable and a missing cable both look “different from normal.”

**Limitations motivating Phase 2.** Four core limitations drove the transition away from PatchCore. First, the memory bank consumes significant RAM and becomes impractical when scaling to all nine connectors. Second, inference is slow because each test image requires millions of patch-to-memory distance computations. Third, It stores features but does not learn a dataset-specific representation, making it sensitive to illumination changes and connector variation. Fourth, the noisy patch-level distance maps produce unstable localization under shadows, reflections or low-contrast conditions.

## A.2 Phase 2: EfficientAD Teacher-Student Framework

The second phase explored EfficientAD [5], a state-of-the-art teacher-student anomaly detection framework that addresses PatchCore’s efficiency limitations while offering sub-5 ms inference latency [5].

**Architecture.** EfficientAD employs two networks. The *Teacher* is a frozen ResNet18 pre-trained on ImageNet, serving as a fixed feature extractor. The *Student* is a randomly initialised ResNet18 with identical architecture, trained to mimic the Teacher’s feature maps on OK images only.

At test time, the squared difference between Teacher and Student feature maps is averaged across channels to produce a low-resolution anomaly map. The maximum value of this map becomes the scalar anomaly score. An additional autoencoder branch targets “logical” anomalies where a part is visually correct but structurally wrong (e.g. a cable routed incorrectly). The published benchmarks report Image-Level AU-ROC of 96.0% and Pixel-Level AU-PRO of 93.3% on MVTec AD and LOCO datasets with approximately 4.5 ms latency per image.

**Training strategy.** All nine connector types underwent standard training of 70 epochs with intermediate checkpoints saved every 10 epochs for retrospective evaluation. Two input modalities were compared: raw RGB (preserving full colour information) and grayscale with CLAHE enhancement (focusing the network on structural differences). Model selection per connector was automated by a two-criterion logic: (1) primary: minimize false positives on the OK test set; (2) secondary: maximize the margin between the worst OK score and the best KO score. Safety thresholds were computed at 95% of the minimum defect score.

**Per-connector results.** The following table summarises the raw RGB modality performance:

| Connector | Modality | Recall | Specificity | FP Rate |
|-----------|----------|--------|-------------|---------|
| conn1     | RGB      | 1.00   | 1.00        | 0.0%    |
| conn2     | RGB      | 1.00   | 0.97        | 3.1%    |
| conn5     | RGB      | 1.00   | 1.00        | 0.0%    |
| conn6     | RGB      | 0.25   | 0.98        | 1.8%    |
| conn7     | RGB      | 1.00   | 1.00        | 0.0%    |
| conn8     | RGB      | 1.00   | 1.00        | 0.0%    |
| conn9     | RGB      | 1.00   | 0.70        | 30.3%   |

Table 3: Performance of initial EfficientAD baseline (Raw RGB). Note the failure modes in conn6 and conn9 which motivated the final Synthetic Negative approach.

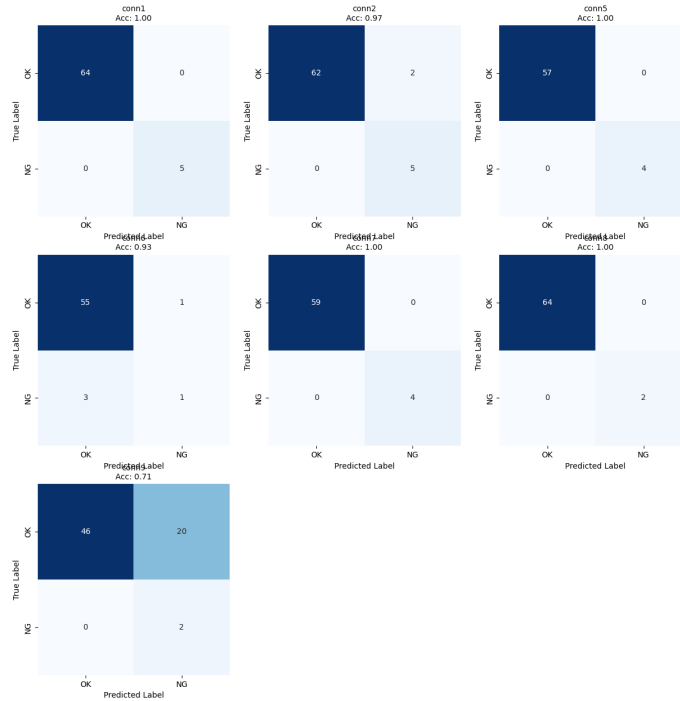


Figure 10: EfficientAD confusion matrices. Perfect diagonal separation on conn1, conn5, conn7 and conn8. Conn9 shows 20 false positives; conn6 shows 3 missed defects.

Connectors 1, 5, 7 and 8 achieved perfect recall with zero false positives. However, two failure modes emerged that proved difficult to resolve within the anomaly detection paradigm.

**Conn6 (low recall, 0.25).** Three out of four defective samples were misclassified as OK. Score distribution analysis revealed that the anomaly scores for these defects were extremely low, overlapping significantly with the normal distribution. This indicated that the Student network had overgeneralised and treated the defect pattern as normal noise. The connector’s PCL region is small relative to the background, so the small defect signal gets lost in the larger non-informative area.

**Conn9 (low specificity, 0.70).** Twenty out of sixty-six OK samples were incorrectly flagged as anomalous. The OK score distribution exhibited a long tail that extended past the safety threshold. Visual inspection suggested that these outliers contained partial occlusions or lighting noise, which the anomaly detector could not distinguish from true structural defects.

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**Multi-step system (Phase 2b).** Building on Phase 2’s insights, a multi-step pipeline was developed (led by Giovanni Passuello) that added a dedicated Occlusion Classifier in front of EfficientAD. The OcclusionCNN is a lightweight three-block convolutional network that classifies each connector crop as OCCLUSION or VISIBLE. The architecture uses Conv2d → BatchNorm → ReLU → MaxPool blocks, expanding from 32 to 128 channels, followed by a fully connected head with 512 neurons and dropout of 0.5.

Only VISIBLE crops proceed to EfficientAD for OK/KO classification. This two-stage design achieved 95.13% overall accuracy on the full 2,853-image dataset, with per-class accuracy of 98.16% for OK, 84.00% for KO and 81.03% for OCCLUSION. The dominant error mode was OCCLUSION samples predicted as KO (56 cases, 11.5% of occlusions), confirming that even with explicit occlusion filtering, partial occlusions visually resemble structural anomalies.

### A.3 Key Insights Carried Forward

Across both phases, several insights directly shaped the final PCL Presence pipeline:

1. **Anomaly detection alone cannot solve the three-class problem.** Both PatchCore and EfficientAD treat everything “different from OK” as anomalous, making it inherently difficult to distinguish missing cables from occluding cables.
2. **Per-connector models are essential.** A shared model trained across all connectors is dominated by the easiest cases (conn3, conn8) and fails on the hard ones (conn6, conn7). Each connector requires its own decision boundary.
3. **The scarcity of real KO data is the binding constraint.** With fewer than 5 defects per connector, no supervised method can learn a robust KO representation directly. The solution must generate training signal from OK data alone.
4. **Occlusions must be handled explicitly.** Relying on the anomaly detector to ignore occlusions produces unacceptable false positive rates. Either a dedicated occlusion classifier or a problem formulation that is inherently robust to occlusions is required.
5. **Continuous scores enable operational tuning.** Both PatchCore and EfficientAD output continuous anomaly scores, enabling threshold-based sensitivity adjustment without retraining. This property was deliberately preserved in the final pipeline.

These findings motivated the shift from anomaly detection to synthetic negative classification described in Section 4, where the problem is reformulated as binary presence detection using inpainting-generated training data.

## B Acknowledgments

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